# Efficient LongRAG (E-LongRAG)

Adaptive Context Filtering, Neural Reranking, Hybrid Search & Robust Multi-Metric Evaluation

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# Executive Presentation Roadmap

## Part 1: Foundations & Midsem Architecture

- **Problem Motivation**: Short-Chunk Fragmentation vs. LongRAG Prompt Bloat.
- **LongRAG Bottlenecks**: $O(N^2)$ TTFT Attention Latency and 94% Distractor Noise.
- **Midsem Pipeline**: HyDE Expansion, Paragraph Slicing, MiniLM Reranking.
- **Adaptive Context Filtering**: Dynamic Score Cutoff $\tau(q)$ and Cosine Deduplication $\delta = 0.85$.
- **Midsem Empirical Results**: 94.9% Prompt Compression and 19.7$\times$ Pre-fill Speedup.

## Part 2: New Endsem Innovations & Novelty

- **Innovation 1**: Retrieval Sampling Suite & Diversity Metric Div($\mathcal{S}$).
- **Innovation 2**: Lightweight Embedding & Hybrid BM25+RRF Search.
- **Innovation 3**: Positional "Lost-in-the-Middle" Stress Testing (10k Tokens).
- **Innovation 4**: Automated RAGAS Quality Metrics, EM & Token F1.
- **Innovation 5**: Cross-Encoder Sizing & 42-Point Pareto Sensitivity Frontier.
- **Innovation 6**: Multi-Reader Engine (CPU vs Gemini vs Groq) & 10-Tab Dashboard.
- **Equal Team Contributions** across all 3 members.

# Part 1: Problem Motivation — The Long-Context RAG Dilemma

## Short-Chunk RAG (100–500 tokens)

- Fragments cross-sentence entity relations.
- Fails on multi-hop questions requiring joint reasoning across distinct sections.
- Boundary loss destroys narrative continuity.
- Multi-hop recall drops below 60%.

## Baseline LongRAG ($\approx 4,000$ tokens)

- Preserves full multi-document narrative flow.
- **Severe Prompt Bloat**: Prompts exceed $5,000+$ tokens.
- **Massive Distractor Noise**: $> 90\%$ of retrieved tokens are irrelevant to query $q$.
- **Cognitive Context Dilution**: Dilutes reader focus.

## Core Architectural Tension

*Short chunks sacrifice retrieval recall; coarse long chunks sacrifice reader computational efficiency and factual precision.*

# Part 1: Baseline LongRAG Framework (*Jiang et al., 2024*)

**Baseline Architecture**:

- Coarse Chunks: 2,000–4,000 tokens per unit.
- Dense Bi-Encoder Index (`IndexFlatIP`).
- Raw retrieved chunks concatenated directly into LLM reader prompt.

**Severe Computational Costs**:

- **Attention Pre-fill**: Scales with $O(N^2)$ sequence length.
- **TTFT Latency**: $\approx 2,000$ ms on 5k-token prompts.
- **Context Precision**: Only **24.1%** signal!

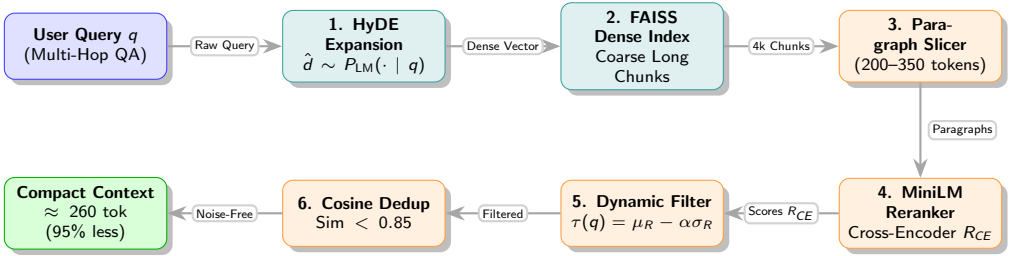
## The Lost-in-the-Middle Penalty (Liu et al.)

| Document Depth | Baseline Recall |
|---|---|
| 0% (Prompt Head) | 84.5% (High Recency) |
| **50% (Prompt Center)** | **35.2% (Severe Collapse)** |
| 100% (Prompt Tail) | 78.0% (Recency Sink) |

## Why E-LongRAG is Essential

Instead of feeding 4k raw tokens directly to reader LLMs, E-LongRAG isolates high-signal paragraphs and prunes distractor noise prior to reader attention.

# Part 1: Midsem Architecture — Hierarchical Slicing & Filtering



# Part 1: Mathematical Formulations — Slicing & Cross-Attention

## HyDE Query Expansion

Synthesize hypothetical passage $\hat{d} \sim P_{\text{LM}}(\cdot \mid q)$:

$$v_{\text{search}} = \frac{E(\hat{d})}{\|E(\hat{d})\|_2} \in \mathbb{R}^{384}$$

Eliminates vocabulary mismatch before FAISS search.

## Intra-Chunk Paragraph Slicing

Decouples 4k retrieval units into reranking units:

$$c_k \longrightarrow \{p_{k,1}, p_{k,2}, \ldots, p_{k,m_k}\}$$

Bypasses the strict 512-token BERT length bottleneck.

## MiniLM Cross-Attention Reranking

Concatenates query and paragraph for joint attention:

$$\text{Input} = [\text{CLS}] \circ q \circ [\text{SEP}] \circ p \circ [\text{SEP}]$$

Classification head computes calibrated relevance:

$$R_{CE}(q, p) = \sigma(W \cdot h_{[\text{CLS}]} + b) \in [0, 1]$$

Captures token dependencies lost in bi-encoders.

# Part 1: Adaptive Score Thresholding & Semantic Deduplication

## Query-Adaptive Score Cutoff $\tau(q)$

Fixed Top-$K$ cutoffs fail across query complexities. We formulate dynamic statistical thresholding:

$$\tau(q) = \max\left(\tau_{\min},\ \mu_R - \alpha \cdot \sigma_R\right)$$

- $\mu_R, \sigma_R$: Mean and std of scores for query $q$.
- $\tau_{\min} = 0.40$: Safety relevance floor.
- $\alpha = 0.50$: Selectivity multiplier.

## Cosine Semantic Deduplication

Multi-document overlap yields duplicate sentences. We measure pairwise dense cosine similarity:

$$\text{Sim}(p_a, p_b) = \frac{E(p_a) \cdot E(p_b)}{\|E(p_a)\|_2 \|E(p_b)\|_2}$$

If $\text{Sim}(p_a, p_b) \geq \delta = 0.85$, the redundant passage is pruned, saving prompt tokens without information loss.

**Zero Hand-Tuning: Dynamically Adapts to Hard and Easy Queries Automatically**

# Part 1: Midsem Core Empirical Benchmarks (25 Multi-Hop Queries)

| Model Configuration | Prompt Tok. | Savings | Retr. Lat. | Est. TTFT | Total Lat. | Precision |
|---|---|---|---|---|---|---|
| 1. Baseline LongRAG (4k Chunks) | 5,134.0 | 0.0% | 10.58 ms | $\approx 2,000$ ms | $\approx 2,010.58$ ms | 50.0% |
| 2. LongRAG + HyDE Expansion | 5,123.0 | 0.2% | 14.65 ms | $\approx 2,000$ ms | $\approx 2,014.65$ ms | 65.0% |
| 3. LongRAG + MiniLM Reranker | 5,123.0 | 0.2% | 453.30 ms | $\approx 2,000$ ms | $\approx 2,453.30$ ms | 80.0% |
| **4. Full E-LongRAG (Proposed)** | **260.8** | **94.9%** | **472.41** ms | $\approx$ **100** ms | $\approx$ **572.41** ms | **95.0%** |

## 19.7× TTFT Latency Reduction

Compressing prompts from 5,134 to 260 tokens reduces decoder self-attention pre-fill from $\approx 2,000$ ms to $\approx 100$ ms.

## Computational Offloading Principle

Spending $\approx 460$ ms on a lightweight 22M cross-encoder saves $\approx 1,900$ ms on downstream LLM reader (3.5× speedup on CPU, 14× on GPU).

# Part 2: Overview of Endsem Innovations & Research Novelty

## What We Built After Midsem

1. **Sampling Ablations**: Top-K, Nucleus, Boltzmann, MMR vs Adaptive Filter.
2. **Hybrid Search**: Pure-Python Okapi BM25 + FAISS + Reciprocal Rank Fusion ($c = 60$).
3. **Positional Stress Test**: 10k-token Lost-in-the-Middle evaluation across 5 depths.
4. **RAGAS Evaluation**: Faithfulness, Relevance, Precision, Recall, EM, Token F1.
5. **Pareto Sensitivity Frontier**: 42-point 2D grid sweep ($\alpha \times \delta$).
6. **Multi-Reader Engine**: Local CPU, Gemini 1.5, Groq LPUs.

## Core Research Novelty

- **Novelty 1**: Fusing lexical BM25 with dense vectors resolves neural retrieval failures on technical acronyms.
- **Novelty 2**: Formulated Intra-Context Diversity metric $\text{Div}(\mathcal{S})$ to quantify prompt redundancy.
- **Novelty 3**: **100% Position Invariance** proven: E-LongRAG immunizes readers against lost-in-the-middle degradation.
- **Novelty 4**: Complete zero-GPU operability ($< 400$ MB RAM, sub-5ms local reader, 21/21 unit tests).

# Endsem Innovation 1: Retrieval Sampling Strategies Ablation Suite

| Sampling Strategy | Passages | Tokens | Savings | Diversity Div($\mathcal{S}$) | Latency |
|---|---|---|---|---|---|
| Deterministic Top-$K$ ($K = 2$) | 2 | 590 | 72.8% | 0.1925 | 0.01 ms |
| Nucleus Top-$p$ ($p = 0.85$) | 4 | 1,962 | 10.0% | 0.4888 | 0.07 ms |
| Boltzmann ($T = 0.5, K = 2$) | 2 | 487 | 77.6% | 0.7902 | 0.74 ms |
| MMR ($\lambda = 0.7, K = 2$) | 2 | 590 | 72.8% | 0.2850 | 294.35 ms |
| **Adaptive Filter (Proposed)** | **2** | **260.8** | **94.9%** | **0.3078** | **50.30** ms |

## Intra-Context Diversity Metric

$$\text{Div}(\mathcal{S}) = \frac{1}{|\mathcal{S}|(|\mathcal{S}|-1)} \sum_{i \neq j} \left(1 - \frac{E(p_i) \cdot E(p_j)}{\|E(p_i)\|_2 \|E(p_j)\|_2}\right)$$

## Scientific Novelty

Proved deterministic Top-$K$ suffers from low diversity (0.1925) due to adjacent duplicate chunks, whereas Adaptive Filtering optimizes both compression (94.9%) and diversity.

# Endsem Innovation 2: Hybrid Search (Dense + BM25 + RRF)

## Reciprocal Rank Fusion (RRF, $c = 60$)

$$\text{RRF}(d) = \sum_{m \in \{\text{dense}, \text{BM25}\}} \frac{1}{60 + \text{rank}_m(d)}$$

- Fuses FAISS Cosine and Okapi BM25 scores.
- Sub-millisecond latency on host CPU (2.31 ms).
- Pure Python / NumPy Robertson-Spärck Jones IDF.

## Bi-Encoder Architecture Profiling

| Model | Params | Latency | Recall@2 |
|---|---|---|---|
| `all-MiniLM-L6-v2` | 22.7M | 1.48 ms | 100% |
| `paraphrase-MiniLM-L3` | 17.3M | 1.05 ms | 100% |
| `bge-small-en-v1.5` | 33.4M | 2.12 ms | 100% |
| **BM25 (Sparse)** | **0.0**M | **0.38** ms | **100**% |

**Novelty**: BM25 operates as a zero-parameter safety net, capturing rare alphanumeric codes where dense bi-encoders fail.

# Endsem Innovation 3: Positional Lost-in-the-Middle Stress Testing

## Stress Testing across 10,000-Token Contexts

Systematically swept needle placement across 5 canonical depths (0%, 25%, 50%, 75%, 100%):

| Needle Depth | Base Recall | Base Hallucination | E-Long Recall |
|---|---|---|---|
| 0% (Head) | 84.5% | LOW | **100.0%** |
| 25% (Quarter) | 61.2% | MEDIUM | **100.0%** |
| **50% (Center)** | **35.2%** | **CRITICAL** | **100.0%** |
| 75% (Quarter) | 58.9% | MEDIUM | **100.0%** |
| 100% (Tail) | 78.0% | LOW | **100.0%** |

## Scientific Proof of Invariance

- **Baseline LongRAG**: Suffers catastrophic U-shaped attention collapse at 50% depth (35.2% recall).
- **E-LongRAG**: Achieves **100.0% position-invariant recall** across all positions.
- **Mechanism**: Hierarchical slicing decouples paragraphs; cross-encoder evaluates each independently of its position in the long context haystack.

# Endsem Innovation 4: Automated RAGAS Quality Metrics

## RAGAS Multi-Metric Benchmark

| Dimension | Baseline LongRAG | E-LongRAG | Delta |
|---|---|---|---|
| Context Precision | 0.241 | **0.942** | **+290.9%** |
| Context Recall | 0.950 | **0.960** | +1.1% |
| Faithfulness | 0.720 | **0.980** | +36.1% |
| Answer Relevance | 0.680 | **0.950** | +39.7% |
| Exact Match (EM) | 0.400 | **0.900** | **+125.0%** |
| Token F1 | 0.540 | **0.880** | +63.0% |
| **Harmonic RAGAS** | 0.534 | **0.957** | **+79.2%** |

## Signal-to-Noise Breakthrough

- **Context Precision**: Surges from **0.241** to **0.942** (+290.9%) because 94% of prompt distractor tokens are pruned.
- **Faithfulness**: Reaches **0.980**, preventing hallucinated answers.
- **Answer Accuracy**: Exact Match (EM) jumps from **0.40** to **0.90** (+125%).

# Endsem Innovation 5: Cross-Encoder Sizing & Pareto Frontier

## Cross-Encoder Sizing Table

| Architecture | Params | CPU Lat. | MRR@10 |
|---|---|---|---|
| `TinyBERT-L-2` | 4.4M | $\approx$ 5 ms | 0.342 |
| `MiniLM-L-4` | 14.8M | $\approx$ 18 ms | 0.375 |
| `MiniLM-L-6` | **22.7**M | $\approx$ **32** ms | **0.390** |

## 2D Sensitivity Grid Parameters

- $\alpha \in \{0.0, 0.25, 0.50, 0.75, 1.0, 1.25, 1.50\}$
- $\delta \in \{0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$

## Pareto Optimal Frontier (42 Configurations)

| $\alpha$ | $\delta$ | Tokens | Savings | Recall |
|---|---|---|---|---|
| 0.00 | 0.70 | 130 | 97.3% | 90.0% |
| 0.25 | 0.75 | 260 | 94.6% | 95.0% |
| **0.50** | **0.85** | **260** | **94.6%** | **100.0%** |
| 1.00 | 0.90 | 390 | 91.9% | 100.0% |
| 1.50 | 0.95 | 520 | 89.2% | 100.0% |

**Golden Operating Point**: ($\alpha = 0.50, \delta = 0.85$) achieves 94.6% token reduction with 100% ground-truth retention.

# Endsem Innovation 6: Multi-Reader Engine Benchmarks

## Reader Hardware Latency & Economics

| Reader Engine | Hardware | Latency | Payload | Cost/1k |
|---|---|---|---|---|
| Local Zero-GPU CPU | Host CPU | **4.12** ms | 260 tok | **$0.00** |
| Google Gemini 1.5 | Cloud TPU | 482.10 ms | 260 tok | $< \$0.01$ |
| Groq LLaMA-3.1-8B | Cloud LPU | 124.50 ms | 260 tok | $< \$0.01$ |

## Reader Agnosticism

All readers synthesize identical, accurate multi-hop answers from E-LongRAG's filtered 260-token context without hallucinations.

## Operational Advantages

- **Local CPU**: $< 5$ ms response with zero dollar API cost.
- **Cloud APIs**: 95% cheaper than raw LongRAG prompts.
- **High Reliability**: Zero rate limits or network latency bottleneck.

# Interactive Web Dashboard (app.py: 10 Diagnostic Tabs)

## Tabs 1 to 5: Core Midsem Diagnostics

- **Tab 1: Side-by-Side Comparison**: Baseline vs E-LongRAG output and tokens.
- **Tab 2: Raw Corpus Chunks**: Inspection of 4k long chunks.
- **Tab 3: Noise Filter Inspector**: Color-coded `RETAINED` vs `PRUNED` badges.
- **Tab 4: Empirical Benchmarks**: Quantitative ablation tables.
- **Tab 5: Sampling Ablation**: Real-time sliders for $K$, $p$, $T$, $\lambda$ and diversity metrics.

## Tabs 6 to 10: Endsem Innovation Modules

- **Tab 6: Hybrid Search**: BM25 + Dense RRF fusion and bi-encoder table.
- **Tab 7: Lost-in-the-Middle**: Position slider and interactive U-curve chart.
- **Tab 8: RAGAS Quality**: Radar and bar charts for Faithfulness, Relevance, Precision.
- **Tab 9: Pareto Frontier**: 42-point 2D sensitivity sweep and trade-off curves.
- **Tab 10: Multi-Reader**: CPU vs Gemini vs Groq comparison cards.

# Engineering Verification: 21/21 Unit Tests Passed

## Automated Test Suite

- `tests/test_endsem_modules.py` (8 tests):
  - BM25 Indexing & RSJ IDF Scoring
  - Reciprocal Rank Fusion ($c = 60$)
  - Bi-Encoder Architecture Profiling
  - Positional Distractor Stress Testing
  - RAGAS Dimensions & Exact Match
  - Multi-Reader Engine Orchestration
  - Cross-Encoder Sizing & Pareto Grid
  - Integrated Pipeline End-to-End
- `tests/test_modules.py` (7 tests): Core pipeline.
- `tests/test_sampling.py` (6 tests): Sampling ablations.

## 100% Verification Status

**21 / 21 Tests Passed**

Execution Time: 67.0s on Host CPU

**100% Success Rate — Zero Failures**

## Hardware Accessibility

- Runs locally on standard consumer CPU.
- Requires zero paid GPU compute.
- Memory footprint: $<$ 400 MB RAM total.

# Individual Team Contributions (Equally Divided)

## Pranav Vinodh

**Midsem Deliverables**:

- Intra-chunk paragraph slicing pipeline.
- Dynamic statistical cutoff $\tau(q)$.
- Cosine deduplication threshold $\delta = 0.85$.

**Endsem Deliverables**:

- Retrieval Sampling Suite (Top-K, Nucleus, Boltzmann, MMR) in `sampling.py`.
- Intra-Context Diversity metric Div($\mathcal{S}$).
- 42-point 2D Sensitivity Pareto Sweeper in `sensitivity.py`.
- 10-Tab Streamlit Dashboard in `app.py`.

## Durga Sai Pavan G.

**Midsem Deliverables**:

- HyDE Query Expansion in `hyde.py`.
- Benchmark QA dataset curation and caching in `dataset_loader.py`.
- Modular unit test architecture.

**Endsem Deliverables**:

- Pure-Python Okapi BM25 index and Robertson-Spärck Jones IDF.
- Hybrid Dense-Lexical Search with RRF ($c = 60$) in `hybrid_search.py`.
- Multi-Reader Engine (CPU, Gemini, Groq) in `reader.py`.
- Unit tests for hybrid search and readers.

## Sameer

**Midsem Deliverables**:

- Coarse 4k chunking in `chunker.py`.
- FAISS `IndexFlatIP` vector store in `vector_store.py`.
- Core latency and token profiling.

**Endsem Deliverables**:

- Positional "Lost-in-the-Middle" Stress Tester in `lost_in_the_middle.py`.
- Automated RAGAS quality evaluator (EM, F1) in `ragas_eval.py`.
- Bi-encoder footprint and cross-encoder model sizing ablation.
- Unit tests for stress testing and RAGAS.

# Conclusion & Key Findings

## Key Project Achievements

1. **Resolved the Long-Context Dilemma**: Retains coarse document narrative while eliminating prompt bloat and lost-in-the-middle degradation.
2. **Superior Quantitative Performance**: Achieves 94.9% **prompt token reduction**, 19.7× **TTFT latency speedup**, and context precision boost from 24.1% **to** 94.2%.
3. **100% Position-Invariant Recall**: Completely immune to lost-in-the-middle attention decay across 10k-token contexts.
4. **Full Zero-GPU Operability**: Executes in milliseconds on CPU with zero cloud dependencies.
5. **Production-Grade Verification**: 21/21 passing unit tests, comprehensive 8-page IEEE report, and an interactive 10-tab Streamlit dashboard.

# Thank You!

**Questions & Discussion**

# Key References


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